

第 2 章 不等式(Inequalities)

2.1 不等式的性质(Properties of Inequalities)

2.1.1 不等关系(Inequality Relations)

定义(Definition): 用符号 “<” “≤” “>” “≥” 或 “≠” 连接两个实数的式子, 称为不等式(inequality)。

示例(Example): $3 > 2$ (3 is greater than 2)、 $2x + 1 \leq 0$ ($2x + 1$ is less than or equal to 0)、 $a \neq 1$ (a is not equal to 1)。

实数大小比较的等价条件(Equivalent Conditions for Comparing Real Numbers):

对任意两个实数 a 、 b (For any two real numbers a and b), 有:

- $a - b > 0 \Leftrightarrow a > b$ (a minus b is greater than 0 if and only if a is greater than b)
- $a - b = 0 \Leftrightarrow a = b$ (a minus b is equal to 0 if and only if a is equal to b)
- $a - b < 0 \Leftrightarrow a < b$ (a minus b is less than 0 if and only if a is less than b)

2.1.2 不等式的基本性质(Basic Properties of Inequalities)

以下性质均基于实数运算规则推导, 适用于所有满足条件的实数 a 、 b 、 c 、 d (All properties are derived from real number operation rules and apply to all real numbers a , b , c , d that meet the conditions):

编号	中文性质描述(Chinese Description)	英文性质描述(English Description)	示例(Example)
1	$a > b \Leftrightarrow b < a$ (对称性)	$a > b \Leftrightarrow b < a$ (Symmetry)	$3 > 2 \Leftrightarrow 2 < 3$
2	$a > b, b > c \Rightarrow a > c$ (传递性)	$a > b, b > c \Rightarrow a > c$ (Transitivity)	$5 > 3, 3 > 2 \Rightarrow 5 > 2$

编号	中文性质描述(Chinese Description)	英文性质描述(English Description)	示例(Example)
3	$a > b \Rightarrow \begin{cases} a + c > b + c \\ a - c > b - c \end{cases}$ (加减性质)	$a > b \Rightarrow \begin{cases} a + c > b + c \\ a - c > b - c \end{cases}$ (Addition/Subtraction Property)	$2 > 1 \Rightarrow \begin{cases} 2 + 3 > 1 + 3 \\ 2 - 4 > 1 - 4 \end{cases}$
4	$a > b, c > 0 \Rightarrow ac > bc$; $a > b, c < 0 \Rightarrow ac < bc$ (乘除性质)	$a > b, c > 0 \Rightarrow ac > bc$; $a > b, c < 0 \Rightarrow ac < bc$ (Multiplication/Division Property)	$a > 2 \Rightarrow 2a > 4$; $a > 2 \Rightarrow -2a < -4$
5	$a > b, c > d \Rightarrow a + c > b + d$ (同向可加性)	$a > b, c > d \Rightarrow a + c > b + d$ (Additivity of Like Inequalities)	$3 > 2, -4 > -5 \Rightarrow 3 + (-4) > 2 + (-5)$
6	$a > b > 0, c > d > 0 \Rightarrow ac > bd$ (同向正数可乘性)	$a > b > 0, c > d > 0 \Rightarrow ac > bd$ (Multiplicativity of Positive Like Inequalities)	$6 > 4 > 0, 3 > 2 > 0 \Rightarrow 6 \times 3 > 4 \times 2$
7	$a > b > 0 \Rightarrow a^n > b^n (n \in \mathbb{N}, n \geq 1, \text{正数乘方性})$	$a > b > 0 \Rightarrow a^n > b^n (n \in \mathbb{N}, n \geq 1, \text{Power Property of Positive Numbers})$	$\frac{1}{2} > \frac{1}{3} > 0 \Rightarrow \left(\frac{1}{2}\right)^3 > \left(\frac{1}{3}\right)^3$
8	$a > b > 0 \Rightarrow \sqrt[n]{a} > \sqrt[n]{b} (n \in \mathbb{N}, n \geq 2, \text{正数开方性})$	$a > b > 0 \Rightarrow \sqrt[n]{a} > \sqrt[n]{b} (n \in \mathbb{N}, n \geq 2, \text{Root Property of Positive Numbers})$	$\frac{1}{4} > \frac{1}{9} > 0 \Rightarrow \sqrt{\frac{1}{4}} > \sqrt{\frac{1}{9}}$

2.2 一元二次不等式的解法(Solution of Quadratic Inequalities)

2.2.1 核心概念(Core Concepts)

- 一元二次不等式(Quadratic Inequality):** 形如 $ax^2 + bx + c > 0$ (或 $\geq, <, \leq$) 的不等式, 其中 $a \neq 0$, a, b, c 为实数(a real number inequality of the form $ax^2 + bx + c > 0$ (or $\geq, <, \leq$) where $a \neq 0$ and a, b, c are real numbers)。
- 判别式(Discriminant):** 对一元二次方程 $ax^2 + bx + c = 0 (a \neq 0)$, 判别式 $\Delta = b^2 - 4ac$, 用于判断方程根的情况(For the quadratic equation $ax^2 + bx + c = 0 (a \neq 0)$, the discriminant $\Delta = b^2 - 4ac$ determines the number of real roots)。
- 求根公式(Quadratic Formula):** 当 $\Delta \geq 0$ 时, 方程 $ax^2 + bx + c = 0$ 的实根为 $x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ (When $\Delta \geq 0$, the real roots of $ax^2 + bx + c = 0$ are $x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$)。

2.2.2 二次函数图像与解集关系(Quadratic Function Graphs and Solution Sets)

一元二次不等式的解集可通过二次函数 $y = ax^2 + bx + c (a \neq 0)$ 的图像直观判断，图像开口方向由 a 的符号决定，根的情况由判别式 Δ 决定。

关键结论(Key Conclusions)

判别式(Discriminant)	$a > 0$ 时图像特征 (Graph Feature when $a > 0$)	解集(Solution Set)
$\Delta > 0$ (两个不相等实根 $x_1 < x_2$)	开口向上，与 x 轴交于 $(x_1, 0)$ 和 $(x_2, 0)$	$ax^2 + bx + c > 0$: $x < x_1$ 或 $x > x_2$; $ax^2 + bx + c < 0$: $x_1 < x < x_2$
$\Delta = 0$ (两个相等实根 $x_1 = x_2 = -\frac{b}{2a}$)	开口向上，与 x 轴相切于 $(x_1, 0)$	$ax^2 + bx + c > 0$: $x \neq x_1$; $ax^2 + bx + c < 0$: 无解(no solution)
$\Delta < 0$ (无实根)	开口向上，与 x 轴无交点	$ax^2 + bx + c > 0$: 全体实数(all real numbers, $\mathbb{R}$); $ax^2 + bx + c < 0$: 无解(no solution)

注：若 $a < 0$ ，先在不等式两边同乘 -1 (不等号方向改变)，转化为 $a > 0$ 的形式再求解

(If $a < 0$, multiply both sides of the inequality by -1 (reverse the inequality sign) to convert it to the form with $a > 0$ before solving).

2.2.3 解法步骤(Solution Steps)

- 标准化(Standardization): 将不等式化为 $ax^2 + bx + c > 0$ (或 $\geq$ 、 $<$ 、 $\leq$)的形式，确保 $a \neq 0$ (Rewrite the inequality into the form $ax^2 + bx + c > 0$ (or $\geq$, $<$, $\leq$) with $a \neq 0$).
- 求判别式与根(Calculate Discriminant and Roots): 计算 $\Delta = b^2 - 4ac$ ，根据 Δ 的值用求根公式求出实根(若存在)(Compute $\Delta = b^2 - 4ac$ and find real roots using the quadratic formula if $\Delta \geq 0$).
- 画图辅助(Graph Assistance): 绘制二次函数 $y = ax^2 + bx + c$ 的大致图像，标注开口方向和与 x 轴的交点(Sketch the graph of $y = ax^2 + bx + c$, marking the opening direction and intersections with the x -axis).
- 确定解集(Determine Solution Set): 根据图像与不等式符号，写出对应解集(Determine the solution set based on the graph and the inequality sign).

可视化图表(Visualization Graphs)

以下是三种判别式情况下二次函数的图像：

图 2.2-1: $\Delta > 0$ 时的二次函数图像($y = 2x^2 - 3x + 1$)

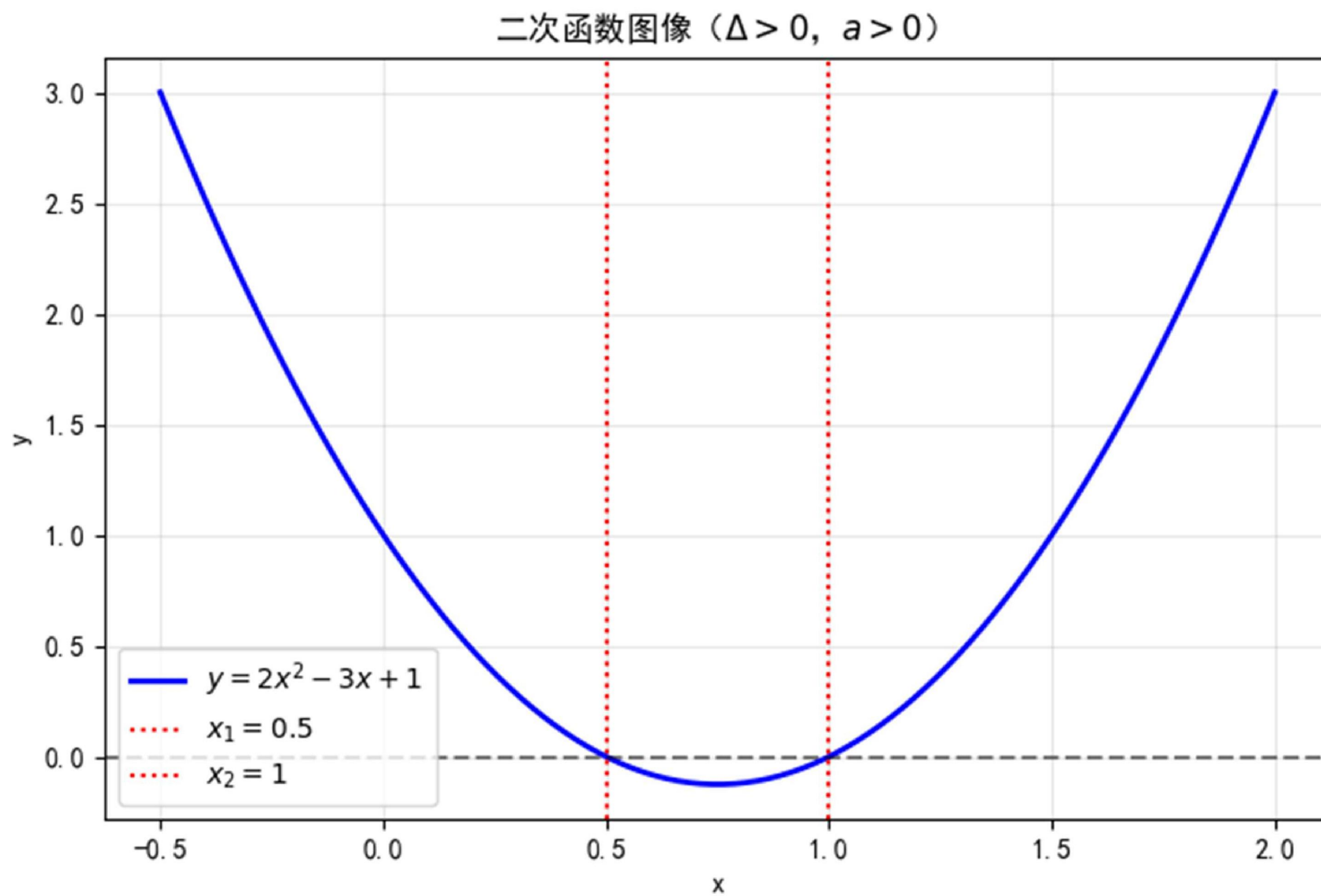


图 2.2-2: $\Delta = 0$ 时的二次函数图像($y = x^2 - 2x + 1$)

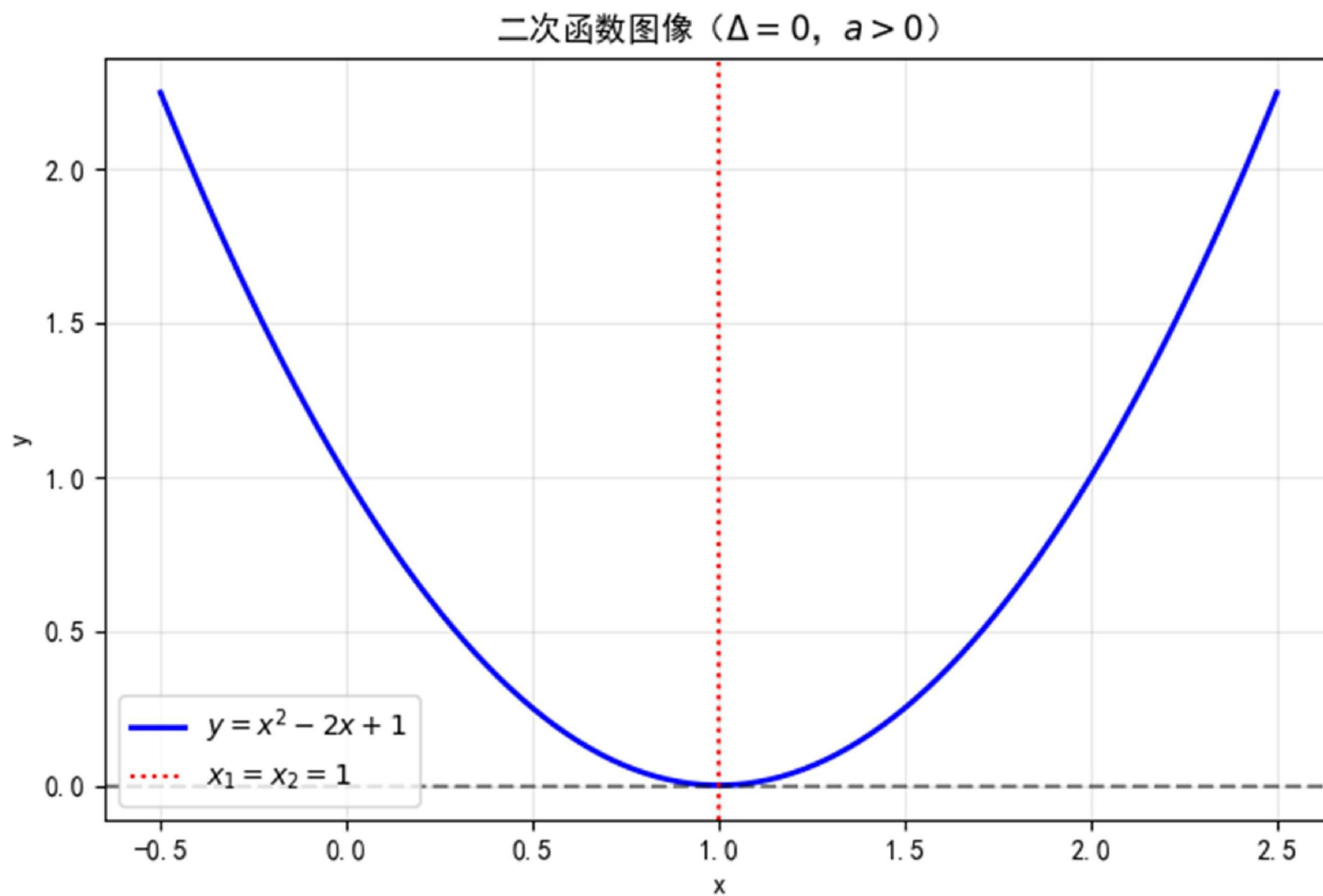
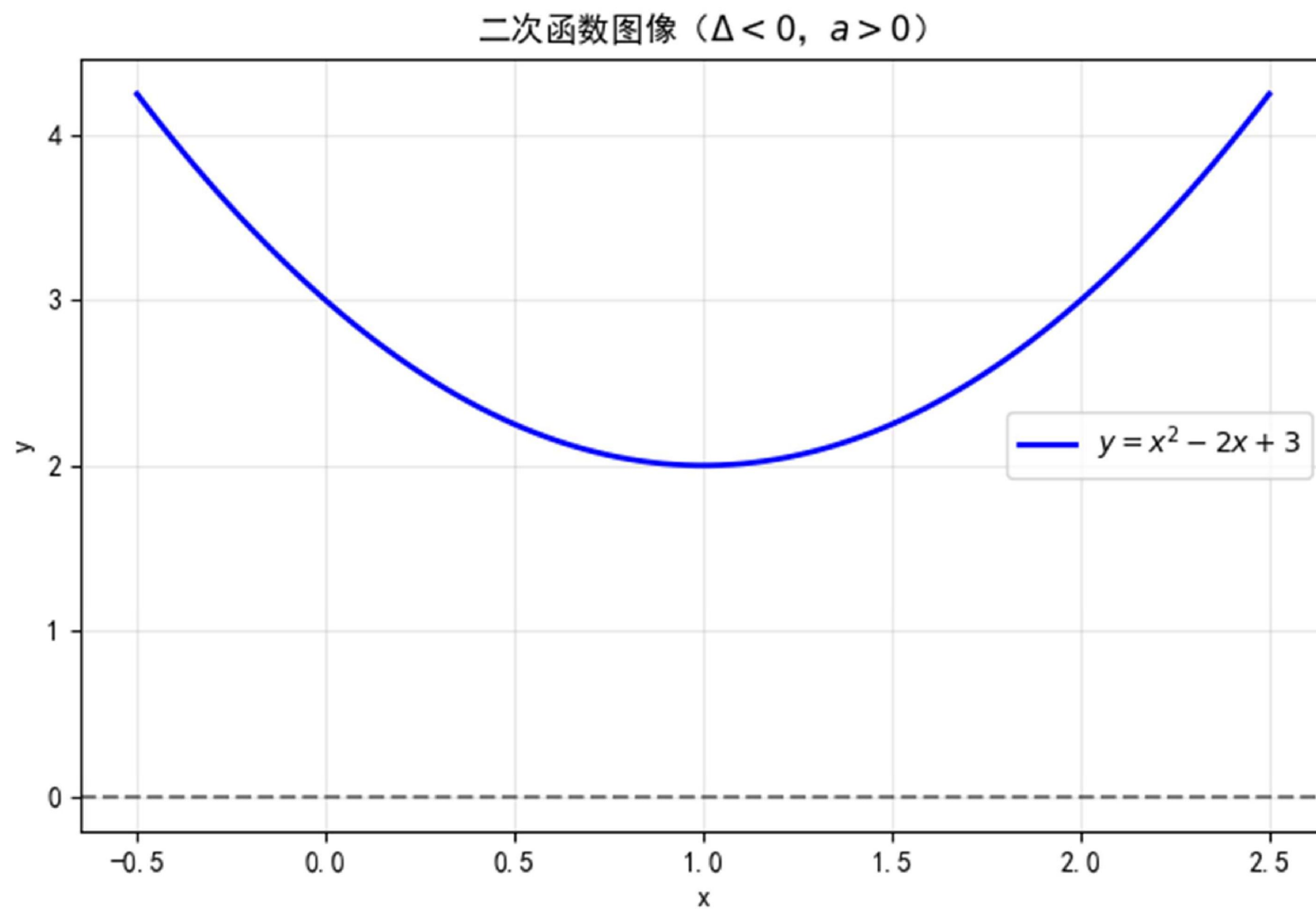


图 2.2-3: $\Delta < 0$ 时的二次函数图像($y = x^2 - 2x + 3$)



2.3 分式不等式的解法(Solution of Fractional Inequalities)

2.3.1 定义(Definition)

分式不等式(Fractional Inequality): 分母中含有未知数的不等式(An inequality with an unknown in the denominator)。

示例(Example): $\frac{x-2}{x+1} > 0$ 、 $\frac{3x+1}{2x-4} \leq 0$; 注意 $\frac{x+5}{2}$ 不是分式不等式(note that $\frac{x+5}{2}$ is not a fractional inequality)。

2.3.2 转化规则(Conversion Rules)

分式不等式的核心解法是转化为整式不等式(不含分母), 需满足分母不为 0 的前提(The core solution of fractional inequalities is to convert them into integral inequalities (without denominators), with the premise that the denominator is not zero):

分式不等式(Fractional Inequality)	等价整式不等式(Equivalent Integral Inequality)
$\frac{f(x)}{g(x)} > 0$	$f(x) \cdot g(x) > 0$ (且 $g(x) \neq 0$)
$\frac{f(x)}{g(x)} < 0$	$f(x) \cdot g(x) < 0$ (且 $g(x) \neq 0$)
$\frac{f(x)}{g(x)} \geq 0$	$\begin{cases} f(x) \cdot g(x) \geq 0 \\ g(x) \neq 0 \end{cases}$
$\frac{f(x)}{g(x)} \leq 0$	$\begin{cases} f(x) \cdot g(x) \leq 0 \\ g(x) \neq 0 \end{cases}$

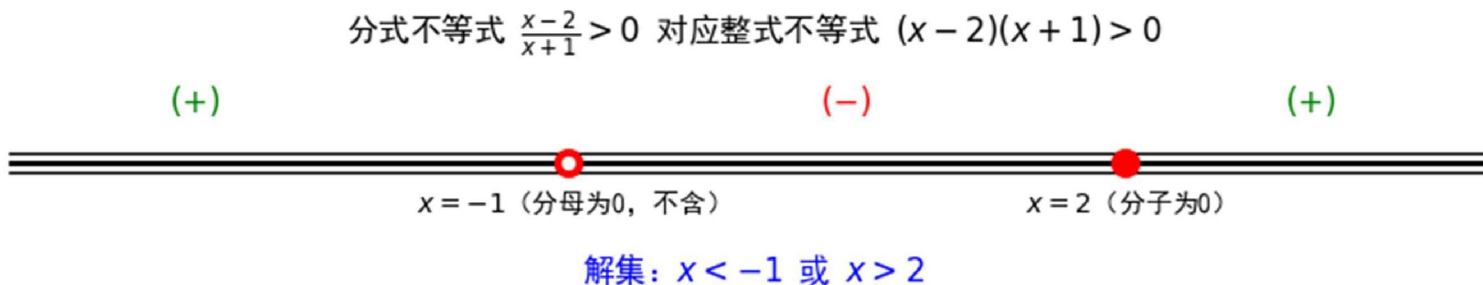
2.3.3 解法步骤(Solution Steps)

- 转化(Conversion): 根据上述规则, 将分式不等式转化为等价的整式不等式(含分母不为 0 的条件)(Convert the fractional inequality into an equivalent integral inequality (including the condition that the denominator is not zero))。

- 求关键值(Find Key Values): 求解整式不等式中对应方程的根, 以及使分母为 0 的未知数的值(These values divide the number axis into intervals)。
- 数轴分区(Number Axis Partition): 将关键值标注在数轴上, 划分出若干区间(Mark the key values on the number axis to divide it into several intervals)。
- 判定符号(Determine Sign): 在每个区间内选取测试点, 判断整式不等式的符号(Select test points in each interval to determine the sign of the integral inequality)。
- 确定解集(Determine Solution Set): 结合符号判定结果, 写出分式不等式的解集(Combine the sign determination results to write the solution set of the fractional inequality)。

可视化图表(Visualization Graphs)

图 2.3-1: 分式不等式数轴分区分析图(以 $\frac{x-2}{x+1} > 0$ 为例)



Summary(总结)

核心要点(Core Points)

- 不等式性质(Properties of Inequalities): 8 个基本性质是变形和求解的基础, 尤其注意乘除负数时不等号方向改变(The 8 basic properties are the foundation for deformation and solution; note that the inequality sign reverses when multiplying or dividing by a negative number)。
- 一元二次不等式(Quadratic Inequalities): 通过判别式判断根的情况, 结合二次函数图像快速确定解集(Use the discriminant to judge the roots and combine the quadratic function graph to quickly determine the solution set)。

3. 分式不等式(Fractional Inequalities): 转化为整式不等式求解, 严格遵循“分母不为 0”的前提(Convert to integral inequalities for solution, strictly following the premise that “the denominator is not zero”).

关键思想(Key Ideas)

- 转化思想(Transformation Idea): 将复杂不等式(分式、高次)转化为简单不等式(整式、一次 / 二次)(Transform complex inequalities (fractional, high-degree) into simple inequalities (integral, linear/quadratic)).
 - 数形结合(Combination of Algebra and Geometry): 利用函数图像直观分析不等式解集(Use function graphs to intuitively analyze the solution set of inequalities).
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